Md Tanvir Parvez

Contact Information

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Research Interests

Developing machine learning models to solve complex real-world challenges, specifically in building scalable solutions using state-of-the-art NLP technologies and data-driven approaches.

Topics: Machine Learning, Natural Language Processing, Data Analysis, Geo-spatial Reasoning

Education

M.Sc. in **Statistics** [Jan 2025 – Jan 2026]
Islamic University (IU), Kushtia, Bangladesh
CGPA: 3.50 on a scale of 4.00

B.Sc. in **Statistics** [Jan 2019 – Jan 2025]
Islamic University (IU), Kushtia, Bangladesh
CGPA: 3.62 on a scale of 4.00 (Last four semesters' CGPA: 3.77)
Position: Ranked 5th in a class of 50 students

Experience

Remote Research Assistant [Mar 2024 – Mar 2025]
Qatar Computing Research Institute, Doha, Qatar

- Contributed to the development of the MapEval benchmark for assessing geo-spatial reasoning in foundation models
- Assisted in data collection, annotation, and evaluation of 28 LLMs with map-based reasoning tasks
- Collaborated with international researchers on publications for top-tier conferences

Technical Skills

[Languages] Python, R, SQL (MySQL), C, LaTeX
[ML/Data Tools] Pandas, NumPy, Scikit-learn, SPSS, Keras, PyTorch, TensorFlow
[Development] JupyterLab, Git, GitHub, Google Colab, VS Code
[Visualization] Power BI, Matplotlib, Seaborn

Publications

1. Mahir Labib Dihan, Md Tanvir Hassan*, **Md Tanvir Parvez***, et al. *MapEval: A Map-Based Evaluation of Geo-Spatial Reasoning in Foundation Models*. ICML 2025 (Spotlight), Vancouver, Canada. (*Equal contribution*)

Research & Projects

Predicting Rice Yields using LSTM Model
Supervisor: Prof. K.S.M. Tozammel Hossain, University of North Texas, USA

- Developed an LSTM model to predict rice yields for three rice types using climate variables
- Achieved R^2 scores of 0.94, 0.81, and 0.94, outperforming baseline models

MapCoder: Multi-Agent Code Generation for Competitive Problem Solving
Supervisor: Prof. Mohammed Eunus Ali, Monash University, Australia

- Helped in running experiments with prompting techniques for LLMs to enhance code generation from natural language
- Received acknowledgment in ACL Conference Paper for contributions to dataset generation

Crops Yield Prediction: Machine Learning Regression Analysis
Supervisor: Prof. Md. Mahabubur Rahman, Islamic University

- Created ML pipeline to predict crop yields using historical climate and agricultural data
- Identified Random Forest as top-performing model through comparative analysis

A Time Series Approach to Short-Term Bitcoin Price Prediction
Supervisor: Prof. Md. Mahabubur Rahman, Islamic University

- Optimized and validated multiple ARIMA configurations, with ARIMA(2,1,2) achieving the best performance, and achieved RMSE of 0.0389 and MAE of 0.0247 on test data

Honors & Awards

- University Merit Scholarship: Islamic University, Bangladesh (2021, 2022)
- Founder, IU Stats for Blood (IUSB) (2023)
- Runner-up, Upazila level Debate Competition (2016)
- Speaker, "Python programming language", Statistics Programming Club, IU (2024)